High-Level Design (HLD)

SailPoint IdentityIQ: HR authoritative source, joiner / mover / leaver and LDAP provisioning

Document	High-Level Design (HLD)
Project	SailPoint IdentityIQ - HR CSV authoritative source, lifecycle and LDAP ApacheDS provisioning
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1. Purpose and scope

This document describes the overall design of a SailPoint IdentityIQ (IIQ) lab implementation that uses an HR CSV file as the authoritative source of identities, calculates a lifecycle status for every person, and grants, changes and removes access in an Apache Directory Server (ApacheDS) LDAP directory through roles. Detailed configuration is in the companion Low-Level Design (LLD).

In scope

- Aggregation of employees, contractors and interns from a pipe-delimited HR CSV file.
- Identity attributes, manager hierarchy and a calculated lifecycle **Status** (PreHire / Active / Terminated).
- Joiner, mover, leaver and rehire handling with email notification.
- LDAP ApacheDS accounts and groups: creation, group membership by birthright and requestable roles, display-name synchronisation, account status read-back and disable / enable.
- Emergency Identity Disable / Enable: an administrator-only quick link that disables or re-enables an identity and its accounts at once, without approval, fully audited.
- Operational tooling: diagnostics, single-account aggregation, catch-up rules.

Out of scope

- Production hardening (SSL/LDAPS, HA, secrets vault), password management and self-service.
- Other target systems (Active Directory, Office 365 itself - Office365Access is an LDAP group only).

2. Solution overview

IIQ reads the HR file, builds an identity for each employee and keeps it current. A global rule works out each person's lifecycle Status from their start and end dates. Roles decide what LDAP access a person should have; IIQ provisions it to ApacheDS. When the Status changes, a lifecycle event runs a workflow that disables or re-enables the LDAP account and sends an email.

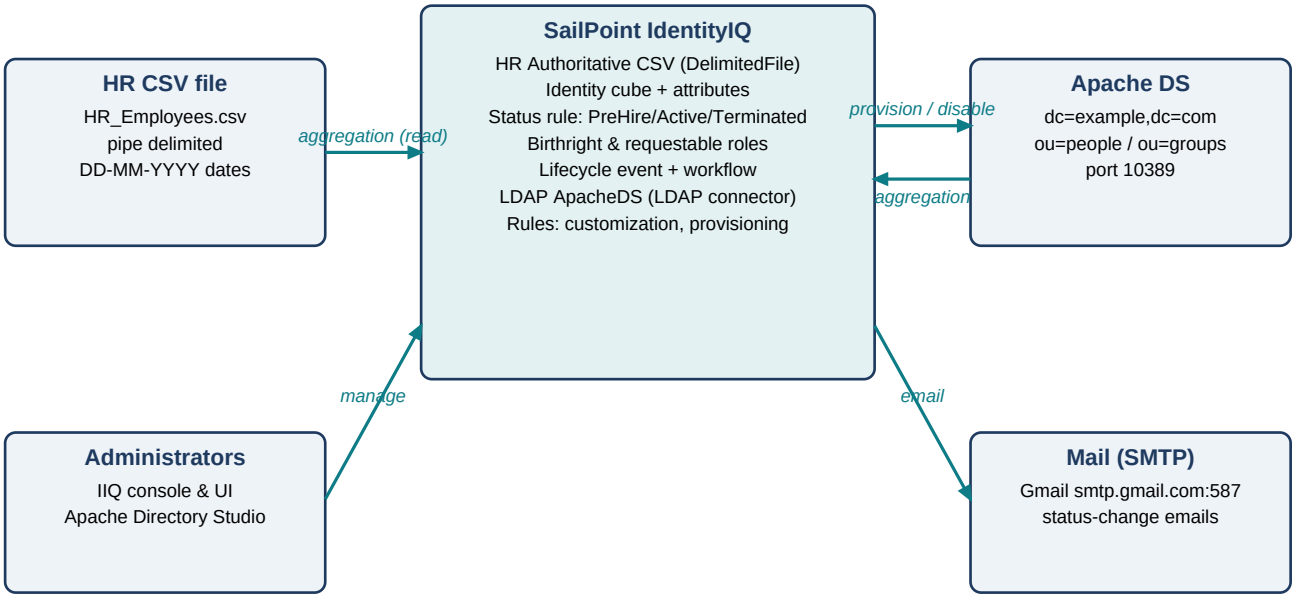


Figure 1 - Logical architecture

3. Systems and components

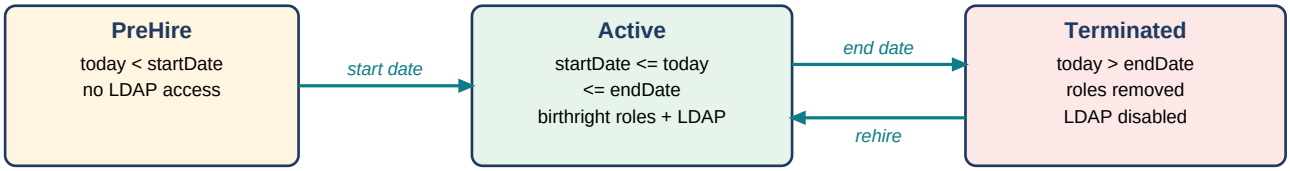
Component	Role	Technology / settings
HR CSV file	Authoritative source of people	C:/SailPoint/hrfeed/HR_Employees.csv, delimiter |, header row, dates DD-MM-YYYY
SailPoint IdentityIQ	Identity governance and provisioning	Tomcat on Windows, http://localhost:8080/identityiq
HR Authoritative CSV	IIQ application for the HR file	DelimitedFile connector, authoritative, account ID = employeeNumber
LDAP ApacheDS	IIQ application for the directory	LDAP connector, localhost:10389, bind uid=admin,ou=system
Apache Directory Server	Target directory	dc=example,dc=com, users in ou=people, groups (groupOfUniqueNames) in ou=groups
SMTP (Gmail)	Notification email	smtp.gmail.com:587 TLS, app password; recipient ashokhearts@gmail.com

4. Identity model

Every HR row becomes one identity named after its **employeeNumber** (for example 1000). Ashok Kumar (1000, now Ashokkumar Chidambaram) is the top manager; every other person reports to him.

Identity attribute	Source
Name (user name)	employeeNumber (set by the Identity Creation rule)
First name, Last name, Manager	HR firstName, lastName, managerId
Display Name	Calculated: firstName + ' ' + lastName (global rule)
Status	Calculated from startDate / endDate (global rule)
Start / End Date, Job Description, Country, Employee Type, Is Manager, Employee Number	HR columns of the same name
Inactive (LCM disabled status)	Calculated: true while Emergency Disable is set, otherwise HR status is not Active (global rule)
Emergency Disable	Set only by the Emergency Identity Status Management Workflow (true / false)
Email	LDAP ApacheDS mail (generated when the LDAP account is created)

5. Lifecycle (joiner / mover / leaver)



Every move between states raises the lifecycle event 'HR Lifecycle - Status Change'
-> workflow: LDAP disable/enable + email to ashokhearts@gmail.com

Figure 2 - Lifecycle states

Event	Trigger	What happens
Joiner	Status becomes Active	Birthright roles assigned; LDAP account created (Create Account policy) and added to groups

Event	Trigger	What happens
Mover	employeeType / isManager / name changes	Roles re-evaluated (groups added or removed); Display Name pushed to LDAP
Leaver	Status becomes Terminated	Roles removed (groups removed); LDAP account disabled; HR account shown disabled; email
Rehire	Terminated -> Active	LDAP account re-enabled; birthright roles assigned again; email
Access request	Office365 Access requested	User added to LDAP group cn=Office365Access
Emergency disable	Administrator, quick link	Emergency Disable = true; identity inactive; LDAP account disabled; audited
Emergency enable	Administrator, quick link	Emergency Disable = false; inactive from HR; LDAP account enabled unless Terminated; audited

6. Access model (roles)

Business role	How assigned	IT role	LDAP group
Birthright - All Active Users	status = Active	LDAP - All Employees	cn=AllEmployees
Birthright - Contractors	Active and employeeType = Contractor	LDAP - Contractors	cn=Contractors
Birthright - Managers	Active and isManager = Y	LDAP - Managers	cn=Managers
Office365 Access	Requestable only	LDAP - Office365Access	cn=Office365Access

7. Integration flows

#	Flow	Direction	Schedule
1	HR account aggregation	HR CSV -> IIQ	After each HR file drop
2	Daily identity refresh (attributes, events, roles, provisioning, attribute sync)	IIQ -> ApacheDS	Daily, just after midnight
3	LDAP account aggregation (accounts, groups, status)	ApacheDS -> IIQ	After refresh / daily
4	Lifecycle workflow (disable / enable, email)	IIQ -> ApacheDS, SMTP	On Status change (during refresh)
5	Emergency Identity Status Management Workflow (disable / enable accounts)	IIQ -> ApacheDS (and other ENABLE-capable apps)	On demand, from the quick link

8. Emergency identity disable / enable

Administrators can cut off a person immediately - for example after a security incident - and restore access later, independently of the HR dates. The quick link **Emergency Identity Disable / Enable** starts the approval-free **Emergency Identity Status Management Workflow**.

Aspect	Design
Who	Only identities with the SystemAdministrator or EmergencyIdentityAdministrator capability see the quick link (DynamicScope); the workflow checks the capability again before changing anything
Form	Identity selector; shows name, status (Active / Disabled), Emergency Disable value and HR status. Only the valid action is offered: Disable for an active identity, Enable for a disabled one. Justification required; requestors cannot target themselves
Identity	Emergency Disable set to true / false; Inactive (LCM disabled status) recalculated at once and kept in step on every refresh by a global rule
Accounts	Disabled / enabled without approval: LDAP ApacheDS by the admin lock, other applications whose connector supports ENABLE through the IIQ provisioner; HR (authoritative) is never changed. Enable never re-enables accounts of an HR-Terminated identity

Aspect	Design
Audit	One AuditEvent per request (also refused ones): requestor, target, action, date and time, workflow result, justification, per-account outcome; plus a server log line

The HR lifecycle respects the flag: a refresh never re-activates an emergency-disabled identity, and a rehire does not re-enable its LDAP account while Emergency Disable is set.

9. Key design decisions

Decision	Reason
Identity name = employeeNumber; correlation on name	Name is always searchable; no extended searchable attribute or schema change needed
Status calculated on refresh, not only on aggregation	People change state on the right day even when the HR file has not changed
LDAP group membership handled by provisioning rules (JNDI)	The LDAP connector sent 'groups' as an attribute; ApacheDS rejected it
Disable by ApacheDS admin lock pwdAccountLockedTime = 000001010000Z	ApacheDS has no enabled flag; the password policy lock is the standard mechanism
Two-level roles (business -> IT)	Business roles cannot hold entitlements; IT roles carry the LDAP group
Setup rules instead of importing ObjectConfig / UIConfig	Importing those objects would overwrite the whole configuration
Inactive calculated by a global rule instead of mapped from HR	Keeps the LCM disabled status in step with Emergency Disable on every refresh
Emergency workflow re-validates admin rights and current state	The quick link scope only hides the link; the workflow is the security boundary
Emergency LDAP disable reuses LDAP ApacheDS - Set Account Status	One proven disable / enable path for leavers and emergencies

10. Non-functional considerations

- **Security:** bind password stored encrypted in IIQ (re-enter it in the UI after import). Lab uses plain LDAP on 10389 and an initial password pattern Welcome@<employeeNumber> - both must change for production.
- **Logging:** log4j2 file appender (C:/SailPoint/logs); rules log with the prefix 'LDAP ApacheDS:' / 'HR CSV'.
- **Performance:** sized for a lab (16 users). Tomcat heap 1-4 GB; LDAP rules reuse one connection per aggregation.
- **Resilience:** catch-up rules (Disable Terminated Users, Aggregate Single Account) recover from missed events.

11. Assumptions, risks and open items

Item	Type	Notes
ApacheDS password policy must be enabled	Assumption	Needed for pwdAccountLockedTime to be accepted and to block logins
LDAP disable / status read-back not yet confirmed in the lab	Open item	Last test: attribute not visible; Studio hides operational attributes by default
Group membership via After Provisioning rule	Risk	Verify uniqueMember changes after first provisioning run
Mail delivery depends on Gmail app password	Assumption	Use the Send Test Email rule to verify
Lifecycle Events page requires Lifecycle Manager	Assumption	Event works without it; view via Debug > IdentityTrigger
Emergency form fields read the selected identity with form.getField()	Open item	Confirm on the first test that the read-only fields fill in after selection
Applications without the ENABLE feature	Risk	Reported as MANUAL by the emergency workflow; disable them on the target system